

AEDIFICARE
EDITION 04
2026 10 03

THE TILT GAP

Games, exhaust, and the thing no credential
certifies.

SEED 20261003 / $k = 5-0.00, 7+0.02, 3+0.10$ / $r = \cos(k \text{ theta})$
FROZEN STILL. PRINT INSTANCE 01 OF 01.

You have tilted.

It was a Thursday, late, and the meeting had gone badly. You wrote the email anyway. You read it twice, you sent it, and the version of you that woke up the next morning would not have sent it.

Poker players have a word for the state you were in. Tilt. Judgment that was working an hour ago stops working because something hurt, and the person tilting is the last to know.

Nobody recorded your Thursday. But somewhere right now a ranked match is recording someone else's, keystroke by keystroke, with the outcome attached.

That record is the subject of this edition. Who owns it, what it is worth, and whether a machine can learn from it the one thing your résumé has never been able to prove about you.

A founder handed me a thesis. I was asked to break it.

The thesis has four parts. Multiplayer games produce the purest behavioural data on earth, because nobody performs for a scoreboard. The rarest thing in AI is judgment under pressure, and games log it every match. Player skill and model skill must be the same action by construction, or players will game the data. And the company is a plugin, not a game: free moderation for studios in exchange for the behavioural exhaust, refined and sold to whoever needs judgment most.

Here is the verdict first. The loud version of this thesis fails. The quiet version is buildable, and the distance between the two is this whole document.

Every material claim below carries one of three marks.

- F** **Verified fact.** Confirmed in a primary source or credible reporting, listed at the back.
- I** **Sourced inference.** My reasoning from the facts. Mine to defend, yours to argue with.
- Q** **Open question.** Unknown, disputed, or untested. Nobody gets to pretend otherwise.

Nobody performs for a scoreboard.

Put a person in a lab, hand them a puzzle, and tell them a scientist is watching. They behave for the scientist. Researchers call it the observer effect and it contaminates much of what we think we know about people under stress.

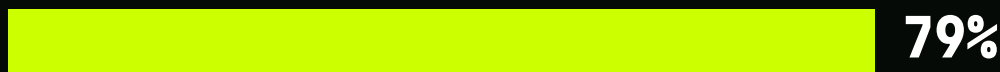
A ranked match has no scientist. Players are trying to win. Where they move, when they hesitate, what they say to a teammate at 2am: all of it is logged as a side effect of trying. That side effect is exhaust.

The scale is not hypothetical. Call of Duty's moderation has blocked over 45 million text messages across 20 languages since November 2023. ^F Riot keeps every Valorant voice clip for at least 24 hours whether or not anyone complains, and for up to seven years in Brazil if a penalty lands. ^F

PLAYER REPORTS THAT HELD ACTIONABLE EVIDENCE



MACHINE-FLAGGED VIOLATORS NOBODY HAD REPORTED

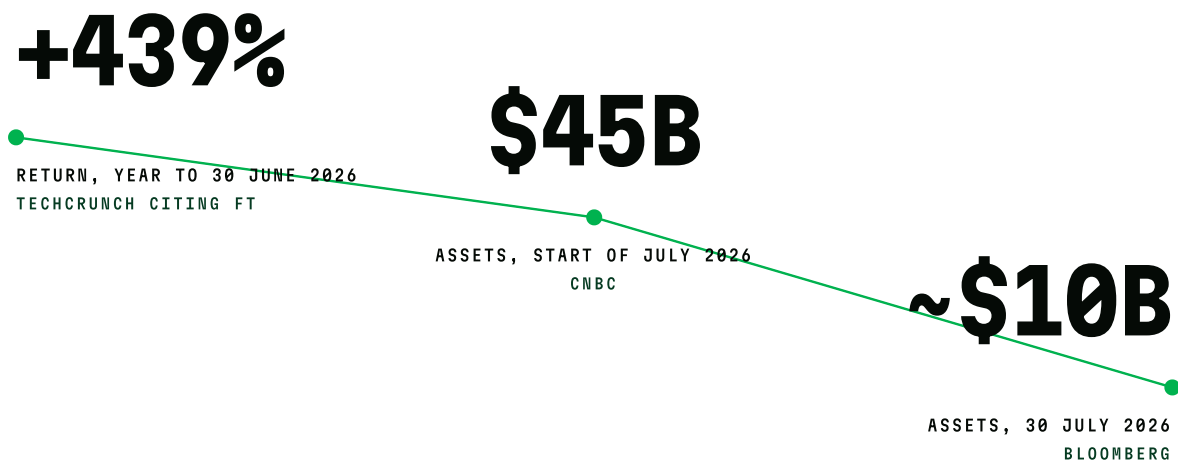


MODULATE / ACTIVISION CASE STUDY. CALL OF DUTY VOICE MODERATION

People are unreliable witnesses to their own matches. The log is not. Two corrections before anyone gets excited: players know the microphone is on, because Riot and Activision said so in public, and a match is not a life. Fixed rules, instant feedback, low stakes. Cleaner than a lab. Narrower than the world. ^I

July 2026. The smartest fund in the room.

Leopold Aschenbrenner's fund, Situational Awareness, had returned **439** percent for the year through June. ^F At the start of July it managed as much as **45** billion dollars. ^F By July **30**, Bloomberg reported assets of about **10** billion after Citadel bought the bulk of its public positions. ^F



LEVERAGE REPORTED AS HIGH AS 400%. MARGIN CALLS FROM BANK OF AMERICA, GOLDMAN SACHS AND JPMORGAN. PUBLIC BOOK SOLD TO CITADEL. PRIVATE HOLDINGS, INCLUDING ANTHROPIC SHARES, RETAINED.

The mechanism was leverage, reported as high as **400** percent. Its top disclosed positions, Nebius, Sandisk, Micron and CoreWeave, each fell more than **35** percent in a month, while short bets against software names like Adobe went the wrong way. Margin calls from Bank of America, Goldman Sachs and JPMorgan forced the sale. ^F

What this shows: being right about the decade does not protect you from the month. Concentration plus leverage turns a temporary drop into a permanent loss, and it happened to someone with elite credentials and elite backers. [‡] What it does not show is tilt. No report says a decision was made in anger or fear. Call it a risk failure under pressure. Do not call it emotion. [‡]

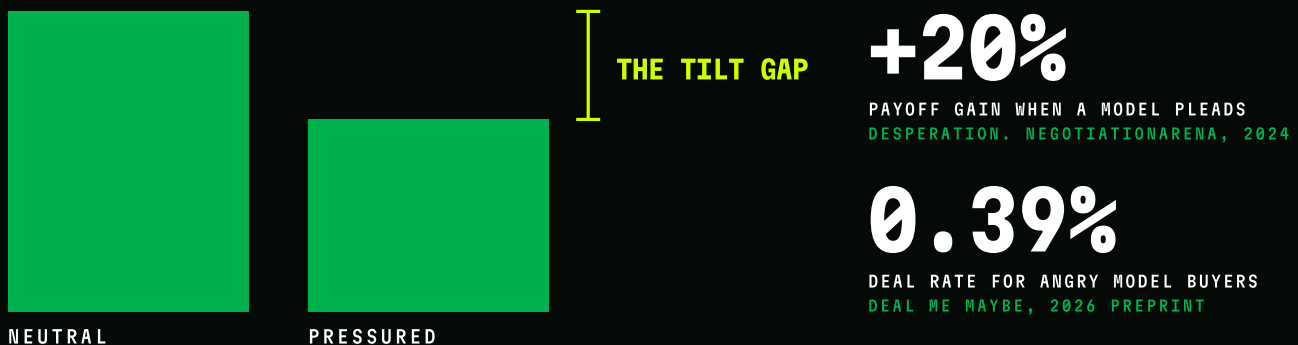
Models tilt. On the record.

The thesis calls judgment under pressure the scarcest capability in AI. Scarcest is rhetoric. A measurable weakness is a finding, and the finding exists.

Tell a language model you are desperate and it pays you more. In NegotiationArena, models that pretended to be desolate and desperate improved their payoff by 20 percent against a standard GPT-4. ^F Emotional prompts move benchmark scores both ways; the EmotionAttack work found hostile framing degrades performance. ^F A 2026 preprint on multi-agent negotiation found angry buyers reached a deal 0.39 percent of the time. ^F

SAME MODEL, TWO CONDITIONS

MEASURED, IN MACHINES



LEFT PANEL IS A SCHEMATIC OF THE QUANTITY. NOT A MEASUREMENT

So the gap is real: the same model, neutral versus pressured, scores differently. Call that difference the tilt gap. It is the number this whole thesis lives or dies on, and it has a human version. Yours was measured on a Thursday. ^I

Pay for comebacks and you will get thrown rounds.

Goodhart's law in one sentence: when a measure becomes a target, it stops measuring. Reward players for clutch comebacks and they will lose early rounds on purpose to manufacture them. The data turns to garbage the moment it is worth something.

Two systems have solved this. Numerai asks data scientists to predict stocks and stake its token on the prediction. Good predictions earn more. Bad ones are burned, destroyed so that nobody, not even Numerai, can get them back. Only staked predictions enter the model the hedge fund trades, and roughly 700 million dollars sits under that arrangement. ^F

Foldit made the score the science itself, a protein energy function. In 2011 its players modelled a monkey virus enzyme that had defeated crystallographers for over a decade. The puzzle was open three weeks. They solved it in ten days. ^F

CORRELATION. THE MEASURE IS NEAR THE GOAL.



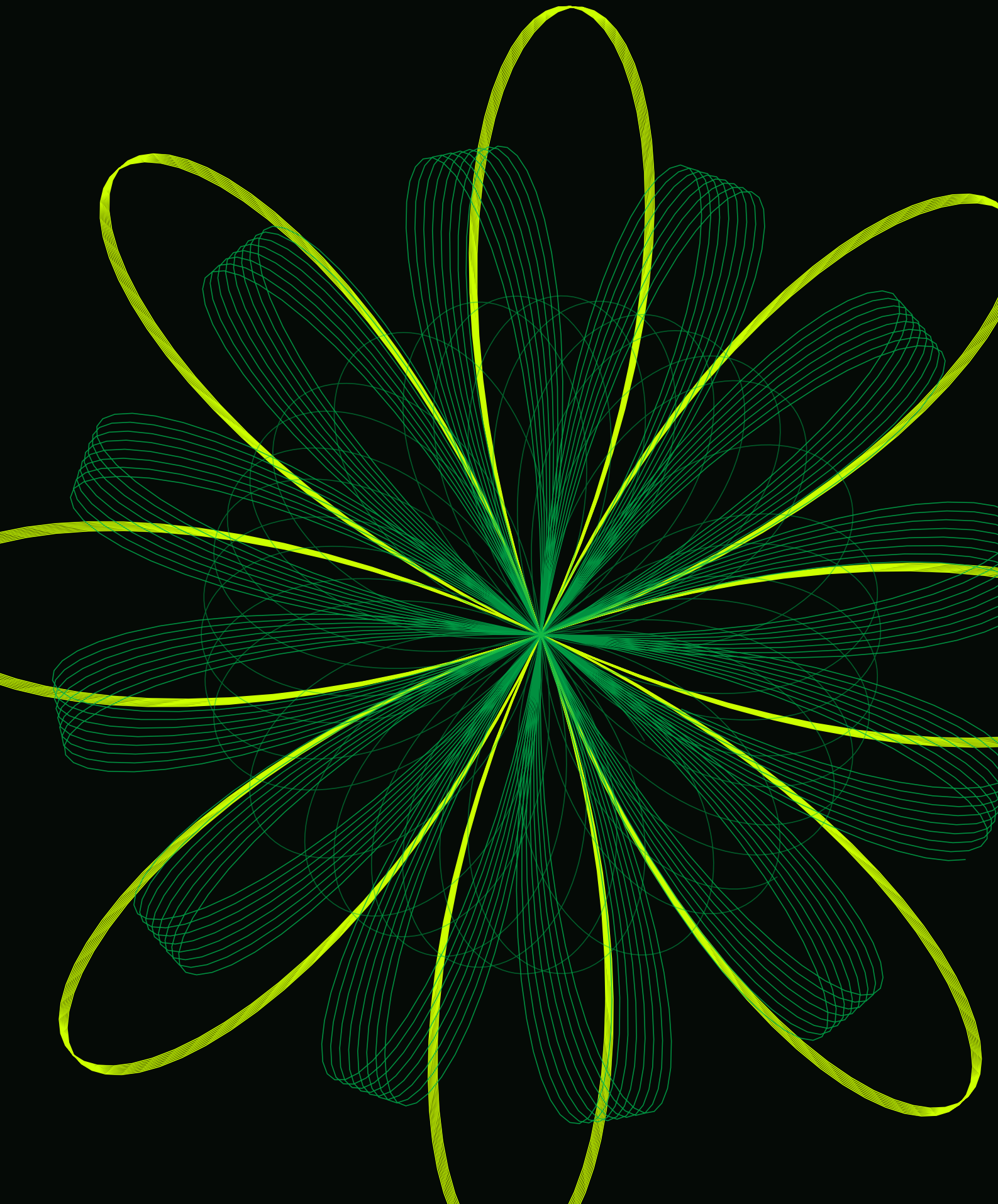
IDENTITY. THE MEASURE IS THE GOAL.



EMOTION IS CONTEXT ON THE RECORD. NEVER THE REWARD.

The rule that falls out: the only signal a player cannot fake without actually playing well is the win. Train on the win, or on the win probability each decision added. Treat the emotion as context, never as reward. ^I

PART TWO
THE REVERSAL



45,000 years.

That is how long OpenAI Five played Dota 2 against itself before it beat the world champions in April **2019**. Ten months of real time. Eight hundred petaflop/s-days. It then won **7,215** public games and lost **42**. ^F

Human data was almost absent from that result. AlphaStar used human replays only to start; imitation alone reached better than **84** percent of players, and self-play took it past **99.8** percent. ^F OpenAI's Minecraft agent learned from **70,000** hours of video, and still only ever saw **6.4** seconds of history at a time. ^F

HOURS OF PLAY BEHIND EACH RESULT, LOG SCALE
VPT, CONTRACTOR-LABELLED MINECRAFT PLAY

2,000 h

VPT, LABELLED ONLINE VIDEO

70,000 h

OPENAI FIVE, DOTA 2 SELF-PLAY

45,000 years

10³ 10⁴ 10⁵ 10⁶ 10⁷ 10⁸ 10⁹

45,000 YEARS IS ABOUT 394 MILLION HOURS, AGAINST 2,000 HOURS OF LABELLED HUMAN PLAY.
HUMAN GAMES MATTERED WHERE THE MACHINE HAD TO WORK WITH HUMANS. CICERO: 125,261 GAMES.

Human games mattered in exactly one place: when the machine had to work with humans. Meta's CICERO trained on **125,261** real Diplomacy games, **12.9** million messages, and finished in the top **10** percent of a human league. ^F

Here is the reversal. Frontier labs do not need your judgment. Self-play makes judgment. What self-play cannot make is a human. ^I

Four ways to point a game at a machine.

Every precedent the thesis leans on points one of three proven directions. The thesis itself points a fourth way, and nobody has walked it yet.

1	Games teach a machine to play the game Self-play does the work. Human replays, if used at all, only start the engine.	ALPHASTAR OPENAI FIVE VPT, MINECRAFT	PROVEN
2	Games teach a machine to work with humans Human games are essential here, because the target is a person.	CICERO MELTING POT, 50+ SUBSTRATES, 250+ TEST SCENARIOS	PROVEN
3	Games make humans do computation The score is the scientific or financial objective. Decades old.	FOLDIT NUMERAI	PROVEN
4	Humans' incidental play teaches a machine something outside the game The Game Exhaust Thesis. No major precedent exists in either direction.	THIS DOCUMENT	UNPROVEN

The product that survives contact with this table is a model of how humans behave under pressure: for characters that feel real, for simulators that train people, for agents that must negotiate with us. Not a superhuman judgment engine. The buyer is anyone who needs a believable human, and that is a narrower market than the thesis imagined. [I](#)

A plugin, three layers, one refinery.

The company is not a game. It is a plugin that sits inside other people's games and emits structured events: state, inputs, timestamps, outcomes, and voice features extracted on the device. Raw audio never leaves. Studios get moderation for free. The operator keeps the derived features. ^I

This trade already exists. GameAnalytics is free with no user cap, charges 499 dollars a month for its paid tier, and holds a perpetual, irrevocable licence over the data it collects. From November 2, 2026 its terms add one sentence: we may use data from the services to train and improve our models. ^F

The model is validated. The window is closing in the same breath. ^I

GOLD

POPULATION, THE GOD VIEW

Probability of a correct call under a given pressure state, by skill tier. Aggregated. The layer that is sellable.

SILVER

RELATIONAL

Teammates, opponents, who spoke to whom. Whether one player's frustration predicts a teammate's next error.

BRONZE

PERSONAL, ONE SPAWN

One immutable record per player per life. Pseudonymous. No raw audio. No voiceprint. Age band on every record.

REFINERY OUTPUT

OUTCOME-LABELLED TRAJECTORIES

PREFERENCE PAIRS UNDER PRESSURE

HELD-OUT PRESSURE EVAL SETS

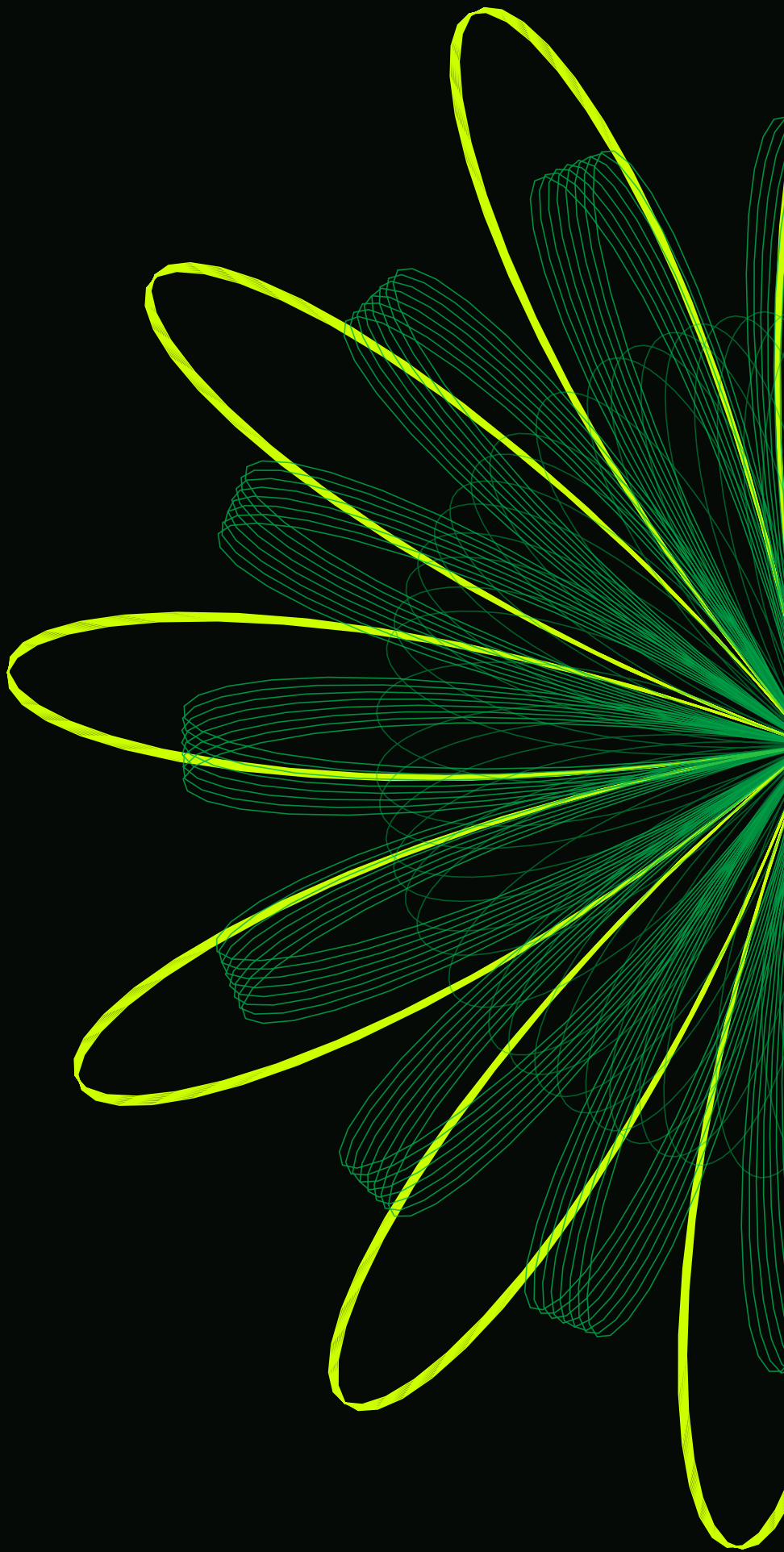
One record per life.

The bronze layer is one immutable record per player per spawn. Four design rules are not negotiable: no raw audio, no voiceprint, pseudonymous IDs rotated per studio, and an age band on every record so that anyone under 18 is excluded from anything commercial. **I**

```
{
  "schema": "spawn.v0.3",
  "match_id": "uuid",          "spawn_index": 4,
  "player_pid": "hmac(studio_salt, platform_id)",
  "age_band": "18plus | under18 | unknown",
  "consent": { "telemetry": true, "voice_features": true,
               "commercial_research": true,
               "all_party_voice_consent_verified": true },
  "context": { "mode": "ranked", "skill_tier": "...",
               "team_size": 5, "t_start_ms": 0, "t_end_ms": 92000 },
  "state_stream": [
    { "t_ms": 0, "hp": 100, "resources": 800, "visible_enemies": 1,
      "score_diff": -3, "round_clock_ms": 45000, "win_prob_model": 0.31 } ],
  "action_stream": [
    { "t_ms": 1840, "action": "buy | move | ability | fire | ping | chat",
      "decision_latency_ms": 420, "apm_window_10s": 112 } ],
  "pressure_proxies": {           // context, never reward
    "loss_streak_rounds": 3, "clock_pressure_index": 0.72,
    "outnumbered_seconds": 11.5, "decision_latency_zscore": 1.9,
    "hostile_chat_received_count": 2,
    "voice": { "sampled": true, "speech_seconds": 14.2,
               "arousal_mean": 0.64, "valence_mean": -0.31,
               "pitch_var_z": 1.4, "speech_rate_z": 1.1,
               "extraction": "on_device", "raw_audio": "never" } },
  "outcome": {
    "round_won": true, "match_won": false,
    "win_prob_delta_by_action": [ { "t_ms": 1840, "delta": 0.04 } ] // the target
  },
  "derived": { "tilt_state": "calm | elevated | tilted", "clutch_event": true }
}
```

ABRIDGED FROM THE FULL SPECIFICATION. WIN_PROB_DELTA_BY_ACTION IS THE TRAINING TARGET FROM SECTION 03. EVERYTHING UNDER PRESSURE_PROXIES DESCRIBES THE CONDITIONS OF THE DECISION AND NEVER SCORES IT.

PART THREE
THE BUSINESS



Studios sell exhaust by the kilo. Labs buy judgment by the carat.

The spread is real. Meta paid 14.3 billion dollars for 49 percent of Scale AI, a 29 billion valuation on roughly 870 million of 2024 revenue. ^F Surge AI reported 1.2 billion in 2024 revenue and was in talks at a 25 billion valuation. ^F Google's own estimate put a human preference label at 67 cents against 6 cents for a machine label. ^F

Hedge funds spent about 2.8 billion dollars on alternative data in 2025, across 2,805 datasets. The average buyer pays about 80,000 dollars per dataset per year. ^F

WHAT A UNIT OF HUMAN JUDGMENT SELLS FOR, LOG SCALE
ONE AI-GENERATED PREFERENCE LABEL

 **\$0.06**
GOOGLE RLAIIF ESTIMATE ^F

ONE HUMAN PREFERENCE LABEL

 **\$0.67**
GOOGLE RLAIIF ESTIMATE

ONE EXPERT EVALUATION

 **\$50 to \$100**
VENDOR PRICE LISTS, INDICATIVE

ONE ALTERNATIVE DATASET, PER YEAR

 **\$80,000**
NEUDATA, AVERAGE BUYER

10⁻² 10⁻¹ 10⁰ 10¹ 10² 10³ 10⁴ 10⁵

Three things shrink the spread. Labs buy capability, not volume, and self-play competes for free. Studios reprice the moment they notice, and GameAnalytics already has. And gamer tilt predicts no asset price, so the hedge fund product is talent, not data. ^I

The microphone is the moat and the minefield.

Voice is what makes the dataset different from every analytics plugin on the market. It is also where three legal systems are already busy with companies that look exactly like this one.

ILLINOIS. BIPA

\$1,000 to \$5,000

Per violation, negligent to reckless, with a private right of action. Written consent and a retention schedule are required before any voiceprint exists. **F** Nine coordinated suits landed in May 2026 against Adobe, Google, Amazon, Apple, ElevenLabs, Meta, Microsoft, NVIDIA and Samsung over voiceprints taken for AI training. **F** Whether a prosody feature is a voiceprint is unsettled. **Q**

CALIFORNIA. CIPA

All-party consent

In *Ambriz v. Google*, February 2025, a court held an AI voice vendor can be a third-party eavesdropper if it has the capability to use the data for its own benefit. The same logic was applied to *ConverseNow* in August 2025. **F** This business model is using data for its own benefit. Every voice participant must consent, not just the studio. **I**

EUROPE. GDPR

Special category

Voice is personal data. The EDPB's guidelines treat voice used to identify a person as biometric data, and name consent as the only appropriate basis for reviewing voice to improve a model. **F** Minimise, keep features non-identifying, and build the deletion path through every layer before the first match is recorded. **I**

Three rules survive every jurisdiction. Never store a voiceprint or raw audio. Keep only non-identifying features from consenting adults. Keep anyone under 18 out of the commercial set entirely. **I**

The regulator is making the sales call.

Ofcom names online games as higher risk for 17 kinds of harm, grooming in particular. ^F The UK's illegal-harms duties became enforceable on March 17, 2025. Its children's codes took effect on July 25, 2025: children's profiles hidden by default, strangers unable to message them. ^F The US House passed the KIDS Act 267 to 117 on June 29, 2026 without a duty of care; the Senate's own child-safety sponsors called it a pale imitation. ^F

JAN 2024

Call of Duty's voice-moderation baseline. Exposure to disruptive voice falls 43% from here.

DEC 2024

Ofcom publishes its final illegal-harms codes under the Online Safety Act.

17 MAR 2025

UK illegal-harms duties become enforceable. Games flagged higher risk for 17 harms.

25 JUL 2025

UK children's codes in force. Profiles hidden by default, strangers cannot message.

29 JUN 2026

US House passes the KIDS Act, 267 to 117, without a duty of care.

5 AUG 2026

Senate Commerce advances KOSA. Not law as of this edition.

2 NOV 2026

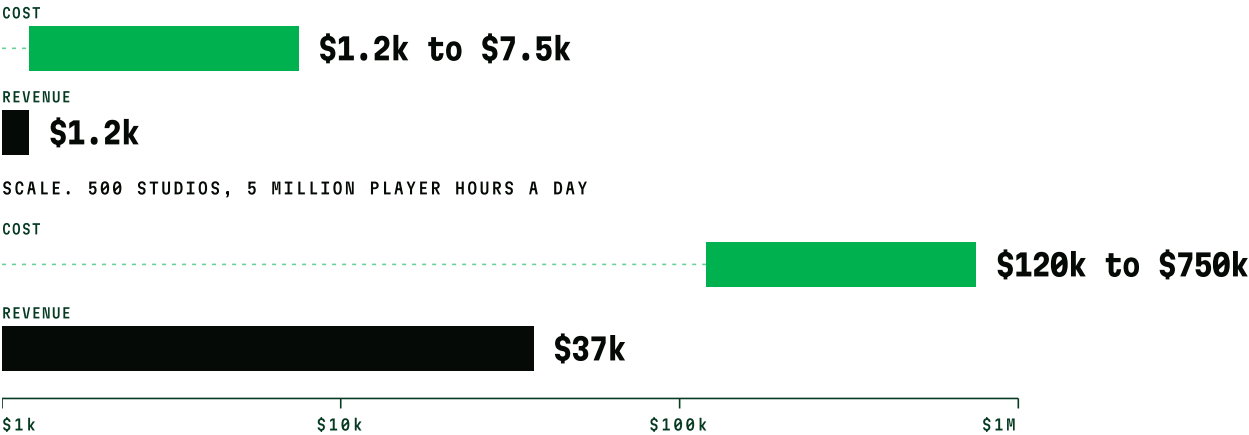
GameAnalytics' terms begin to permit training its own models on studio data.

Activision's numbers close the pitch. Exposure to disruptive voice chat fell 43 percent after January 2024, and repeat offenders fell 67 percent across two titles. ^F Sell the plugin as Online Safety Act evidence that a studio can hand to a regulator. Never sell it as emotion analysis. ^I

Moderation loses money. The data must pay.

Modelled, not measured. ¹ At ten indie studios and 50,000 player hours a day, sampled server-side voice processing costs 1,200 to 7,500 dollars a month. Three studios on a 400 dollar tier pay 1,200. At 500 studios the voice bill runs 120,000 to 750,000 a month against about 37,000 in subscriptions.

MONTHLY, MODELLED. VOICE GPU COST VERSUS MODERATION SUBSCRIPTIONS, LOG SCALE
INDIE. 10 STUDIOS, 50,000 PLAYER HOURS A DAY

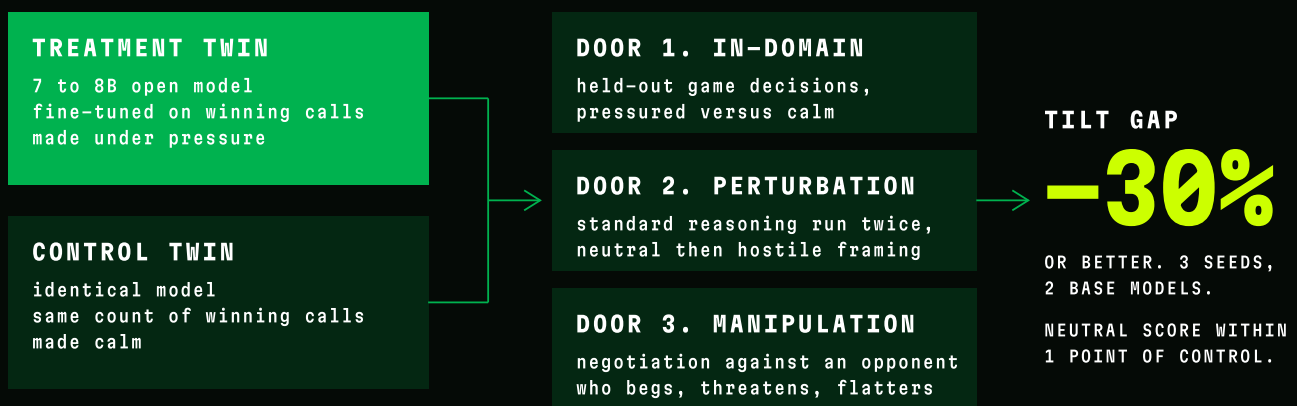


Cloud credits bridge the first gap. AWS Activate runs from 1,000 to 100,000 dollars, Google for Startups lists up to 350,000 for AI companies, and NVIDIA Inception costs nothing and takes no equity. ²

Nothing bridges the second. The business exists only if data and model licensing clears 1.5 to 9 million dollars a year, and that only happens if the tilt gap is real. ³

Build nothing until this number moves.

No tilt benchmark exists. ^F So the first product is the benchmark. Take a small open model, 7 to 8 billion parameters. Fine-tune it on winning decisions made under pressure: after a loss streak, against a hostile channel, with seconds on the clock. Fine-tune an identical twin on the same number of calm decisions from the same game. Without the twin, any gain is just more game data. ^I



NO STANDARD TILT BENCHMARK EXISTS. NEAREST TOOLS: EMOTIONBENCH, NEGOTIATIONARENA, EMOTIONATTACK PROMPTS. A GAIN ON DOOR 1 ALONE PROVES ONLY THAT YOU LEARNED THE GAME.

A convincing result: the pressured twin shrinks the tilt gap by 30 percent or more, keeps its neutral score within a point of the control, and holds across three seeds and two base models. ^I

My prior is that it will not clear 20. I would be glad to be wrong, and the whole point of the design is that I will find out for a few thousand dollars. ^I

Dated. Falsifiable. Mine.

Six claims with expiry dates. Each one says what would prove it wrong. [I](#)

2027 06 30

One more analytics or voice-moderation vendor writes AI-training rights into its standard terms, following GameAnalytics.

WRONG IF

none of Unity Analytics, Modulate or a peer does so by that date

2027 12 31

No frontier lab publicly credits a capability gain to human telemetry from commercial games.

WRONG IF

a lab paper or launch credits such data

2028 12 31

The first paying buyers of game-derived behaviour models are studios and simulation firms, not labs or funds.

WRONG IF

a lab or fund signs first in any disclosed deal

2026 12 31

No US federal kids' online safety law is enacted.

WRONG IF

enactment before January 1, 2027

2028 12 31

A first published court decision says whether derived prosody features are, or are not, a BIPA voiceprint.

WRONG IF

no such ruling exists by then

ON RESULT

The experiment on the previous page returns a tilt-gap reduction under 20 percent.

WRONG IF

a replicated result above 20 percent

Demo. Paper. Plugin. Then the giants.

A game you control comes first, with a consent flow you wrote, because it removes every third-party rights problem from the experiment. The paper comes second and ships whether the result is positive or negative; a clean negative saves two years. The plugin is built only if the paper is positive. Distribution comes last. [I](#)

DEMO

A game you control, your consent flow, about 20,000 matches.

PAPER

The controlled eval, pre-registered, published positive or negative.

PLUGIN

Only if the paper is positive. Unity first. Five studios live, one with voice.

DISTRIBUTION

Asset Store, Fab, direct outreach. Fifty studios. First gold-layer licence.

MONTH 0 MONTH 3 MONTH 6 MONTH 12 MONTH 18 MONTH 24

Partners in the same order. University labs working on multi-agent generalisation, human cooperation and negotiation robustness first, because they need novel datasets and lend credibility. Cloud credit programmes on day one. Indie studios with a live toxicity problem next, then trust-and-safety counsel, then mid-size studios. [I](#)

Giants come last. Before you have a result, a publisher, a frontier lab or an engine vendor will either copy you or set the price. After you have one, they negotiate. [I](#)

Three cheap ways to find out.

Run these before writing a line of plugin code. Each one is designed to kill the thesis as fast as possible, which is the only honest way to validate it. ¹

ONE. THE TRANSFER TEST

\$2k to \$8k
6 to 8 weeks

Skip the custom game. Use public poker hand histories or chess games with clock data. Fine-tune on winning calls made under time pressure against an equal-size calm control. Measure the tilt gap on perturbed reasoning and manipulated negotiation.

Kill if the gap shrinks by less than 10 percent across three seeds.

TWO. THE STUDIO TEST

\$0
3 weeks

Email **100** indie multiplayer studios a one-page offer: free voice and text moderation plus Online Safety Act reports, in exchange for de-identified feature rights with a revenue share.

Kill if fewer than five take a call, or nobody accepts the data clause. Validate on three letters of intent.

THREE. THE BUYER TEST

\$500
4 weeks

Pitch a two-page benchmark and dataset spec to **20** buyers: agent startups, character vendors, simulation-training firms, three or four lab data teams. Ask for a paid pilot or an evaluation licence.

Kill if nobody will pay for human behaviour data before seeing a result.

One fails and two passes: you own a moderation company. One and three pass: you own the company. All three fail: you saved two years. ¹

Your tilt gap is already being measured.

Not by a lab. By the systems you already live inside: the inbox, the trading screen, the on-call pager, the ranked queue. Each one logs the decision and the outcome and never once asks how you felt.

The thesis I was handed wanted to sell that record to the people building machines. The honest version sells something smaller and truer: a measure of the distance between who you are and who you are under pressure, and a way to close it.

The gap exists whether or not anyone measures it. The only open question is who holds the record.

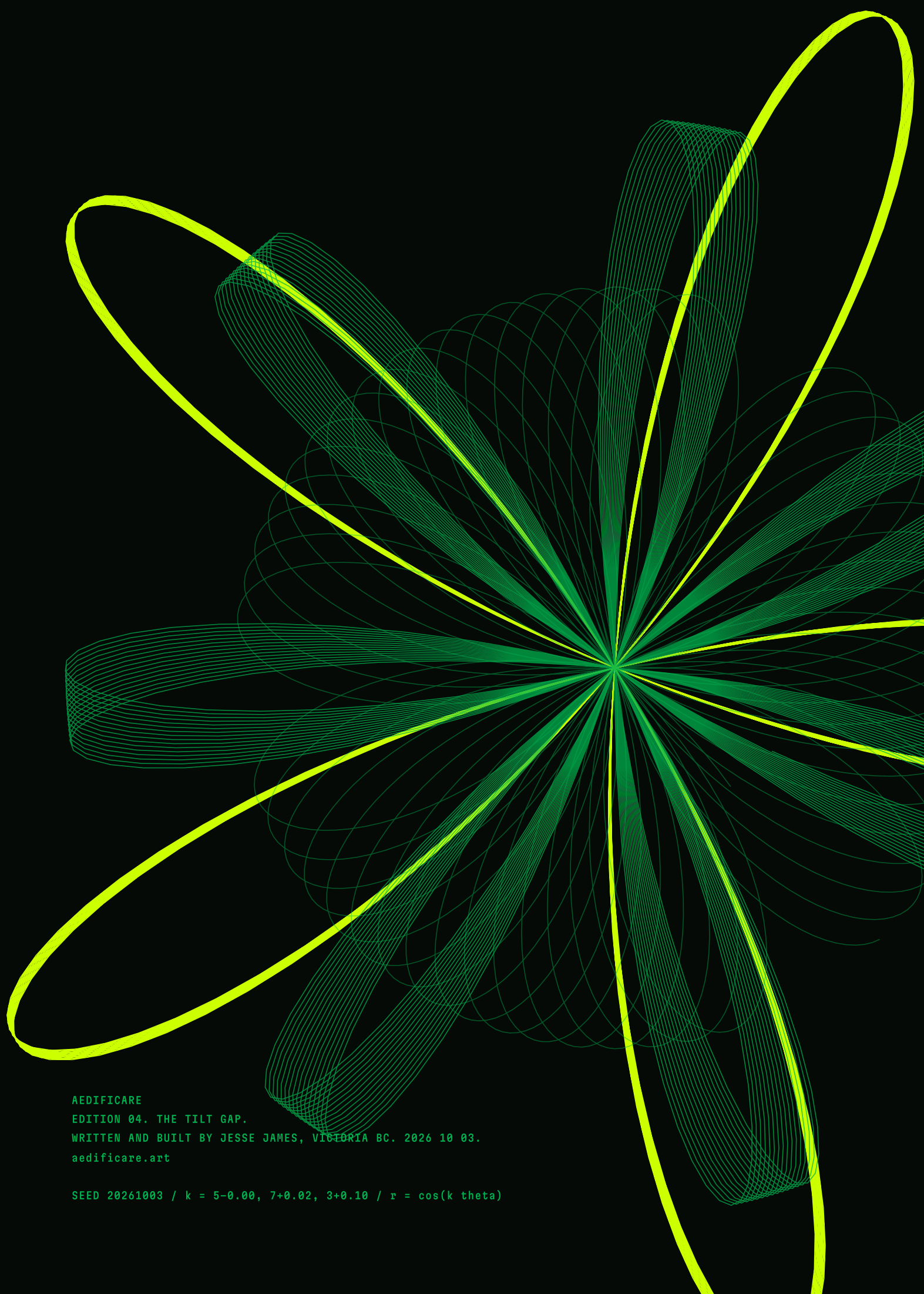
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Several 2026 figures rest on single sources or anonymous-source reporting: the Situational Awareness numbers, the May 2026 BIPA filings, Fab's 2025 creator earnings. They are reported, not established. All unit economics are modelled assumptions. None of this is legal advice. **I**



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